

An Exploration of Annotated Transposon Sequences in *C. elegans*

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Introduction and Overview

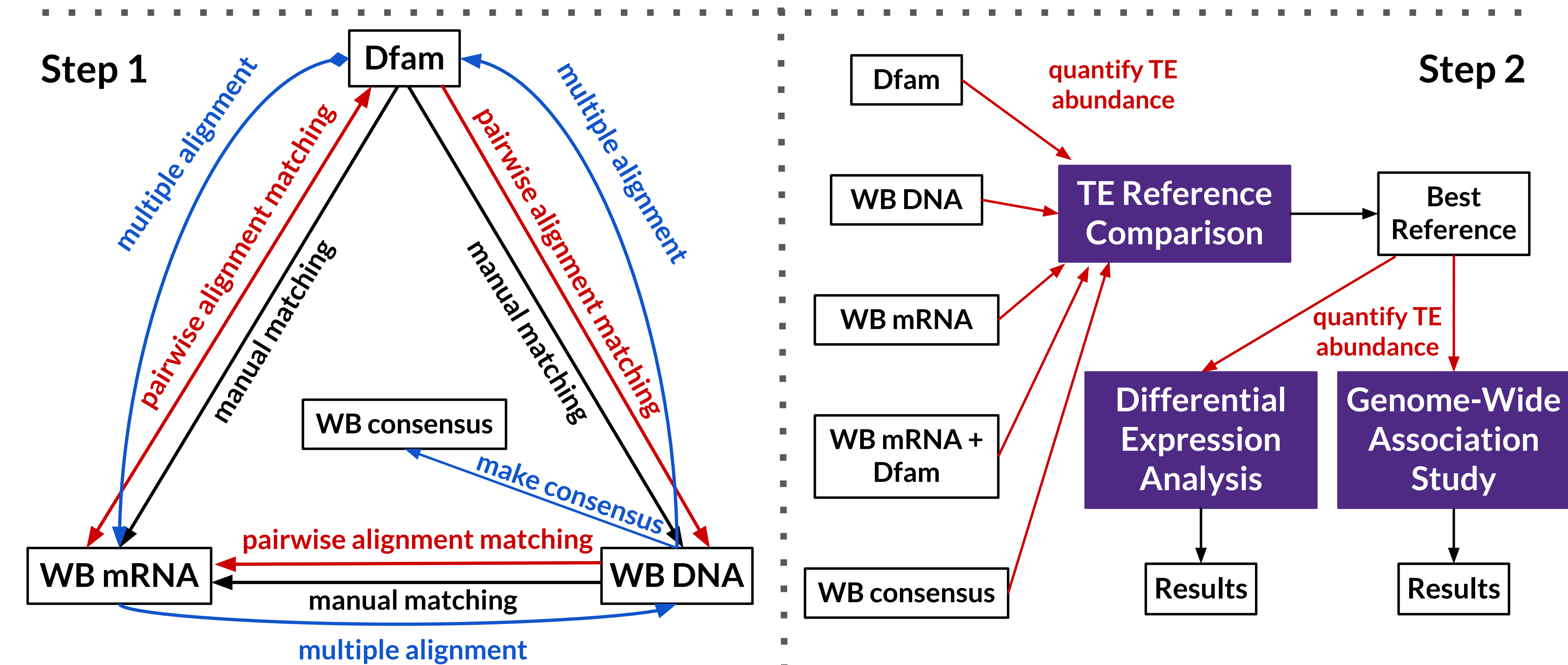
Transposable elements (TEs), also known as transposons, are mobile segments of DNA that can migrate from one part of the genome to another through either an RNA intermediate or a cut-and-paste mechanism. They can be grouped into families which share similar characteristics (Bessereau 2006). Transposons can contribute to genomic variation and evolution by inserting into coding sequences or regulatory regions of the genome, potentially altering an organism's phenotype (Laricchia *et al.* 2017).

We can study natural variation in TEs across *C. elegans* strains through quantification of RNA-seq data (Laricchia *et al.* 2017). This requires us to provide a reference list of annotated transposon sequences with which mRNA transcripts can be matched. We therefore turn our attention to existing online databases that describe *C. elegans* TE sequences. The WormBase (WB) online database provides annotated sequences of 11257 *C. elegans* TE DNA sequences—of which 761 are known and 10496 are predicted—as well as annotated sequences of 366 *C. elegans* TE mRNA transcripts. The Dfam online database provides a list of 187 TE DNA consensus sequences associated with *C. elegans*—one per TE family. The WormBase and Dfam databases both contain information regarding the transposon family classification of each sequence.

We aim to (i) compare three lists of TE sequences (WB DNA, WB mRNA, Dfam DNA), (ii) determine which list serves as best reference for quantification, and (iii) demonstrate methods of analyzing quantification data.

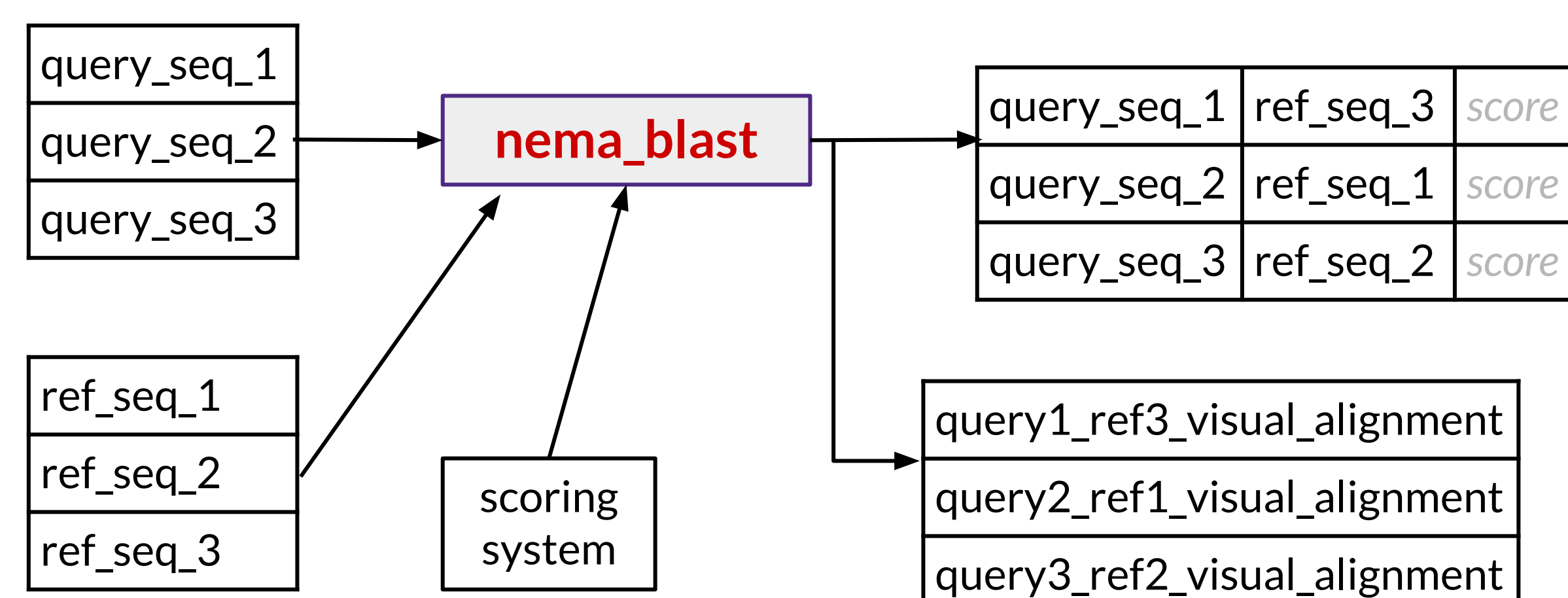
Materials and Methods

WormBase transposon DNA sequences ("WB DNA"), WormBase transposon mRNA transcript sequences ("WB mRNA"), associated transposon annotations, as well as the *C. elegans* reference genome and annotations were obtained from WormBase release WS276. Dfam transposon DNA consensus sequences ("Dfam") and associated annotations were obtained from Dfam release 3.6. Gene sequence variations were obtained from CeNDR release 20200815. Packages kallisto/sleuth (Bray *et al.* 2016), PEmRNA-seq-nf, and NemaScan were used to quantify abundances of transcripts, analyze differential expression, and perform GWA analysis. RNA-seq data on *C. elegans* strains and wild isolates were sourced from the Andersen Lab.



Pairwise Alignment Matching

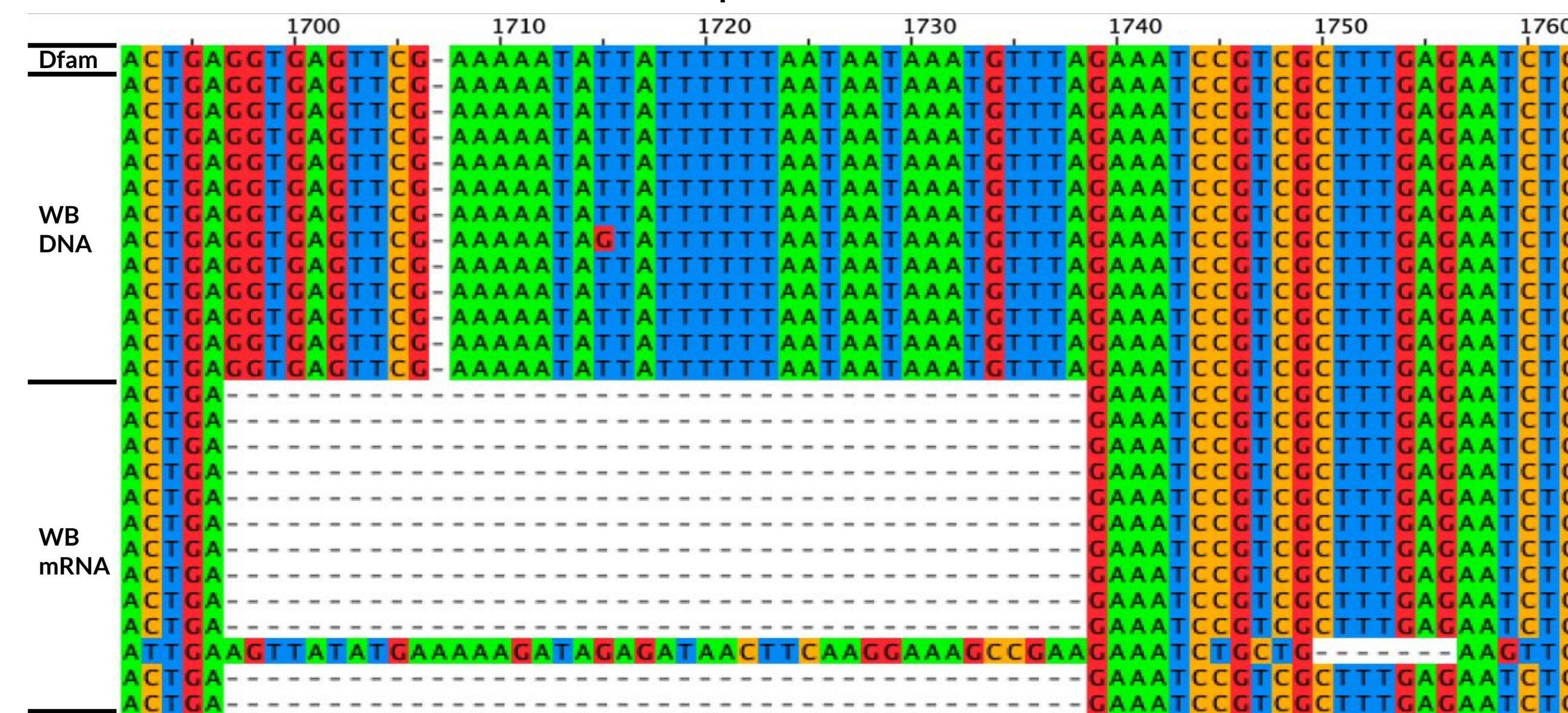
nema_blast is an automatic pipeline for matching a list of query sequences to a list of reference sequences and their reverse complements using a custom local/global alignment algorithm.



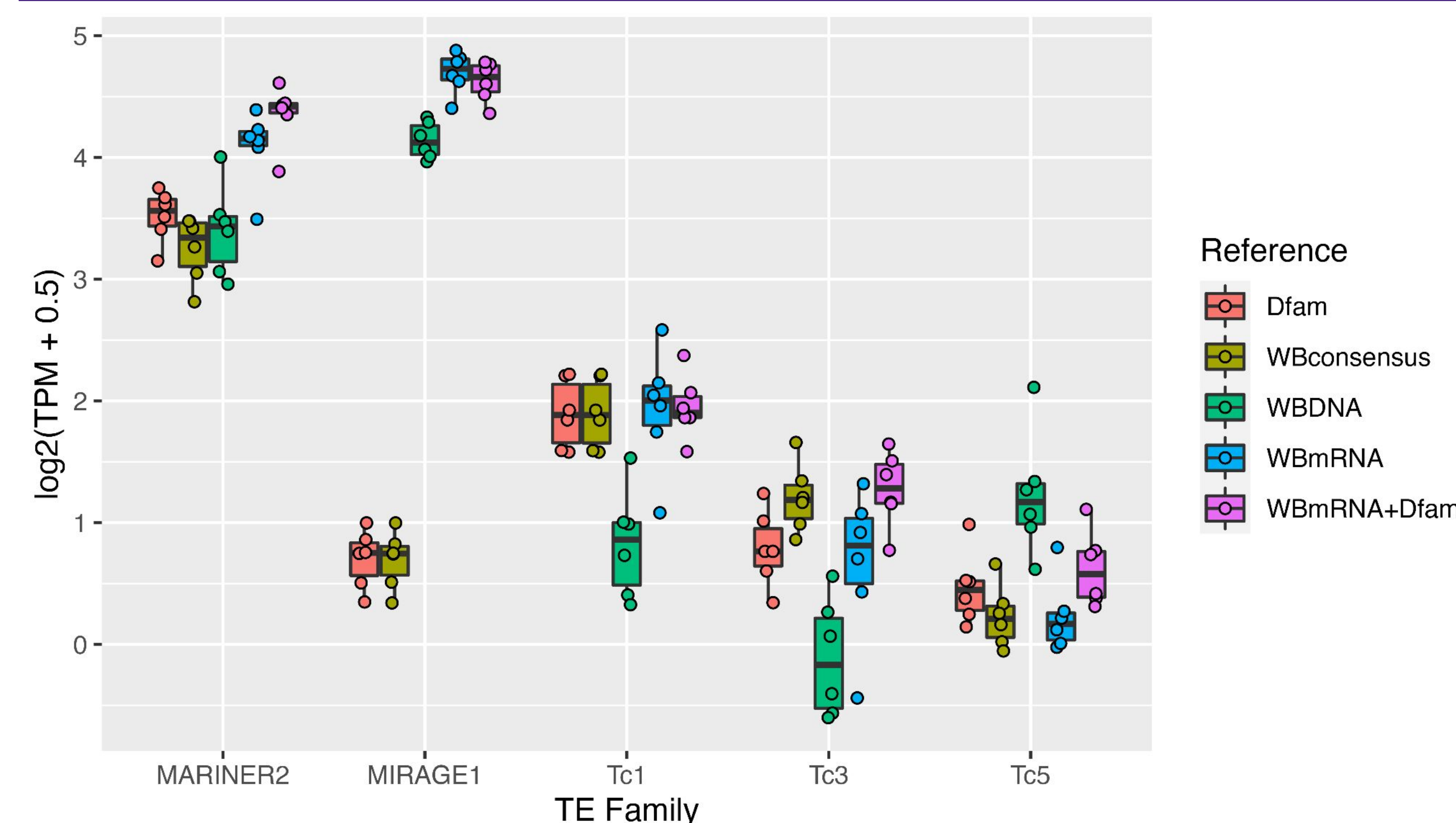
We used **nema_blast** to match WB mRNA transcripts to Dfam sequences (see **match_data**). **nema_blast** verified manually matched WB mRNA–Dfam pairs with an accuracy level of ~99.5%.

Multiple Alignment + Consensus Generation

To visualize differences between TE sequence lists for each TE family, we developed **many_multiple_align**, a pipeline that calls the MUSCLE alignment algorithm (Edgar 2004) to rapidly align many lists of multiple sequences. This pipeline also generates consensus sequences; we used this feature to produce consensus sequences for each WB DNA TE family. A portion of the alignment of several WormBase sequences associated with the Dfam Tc1 sequence is shown below.

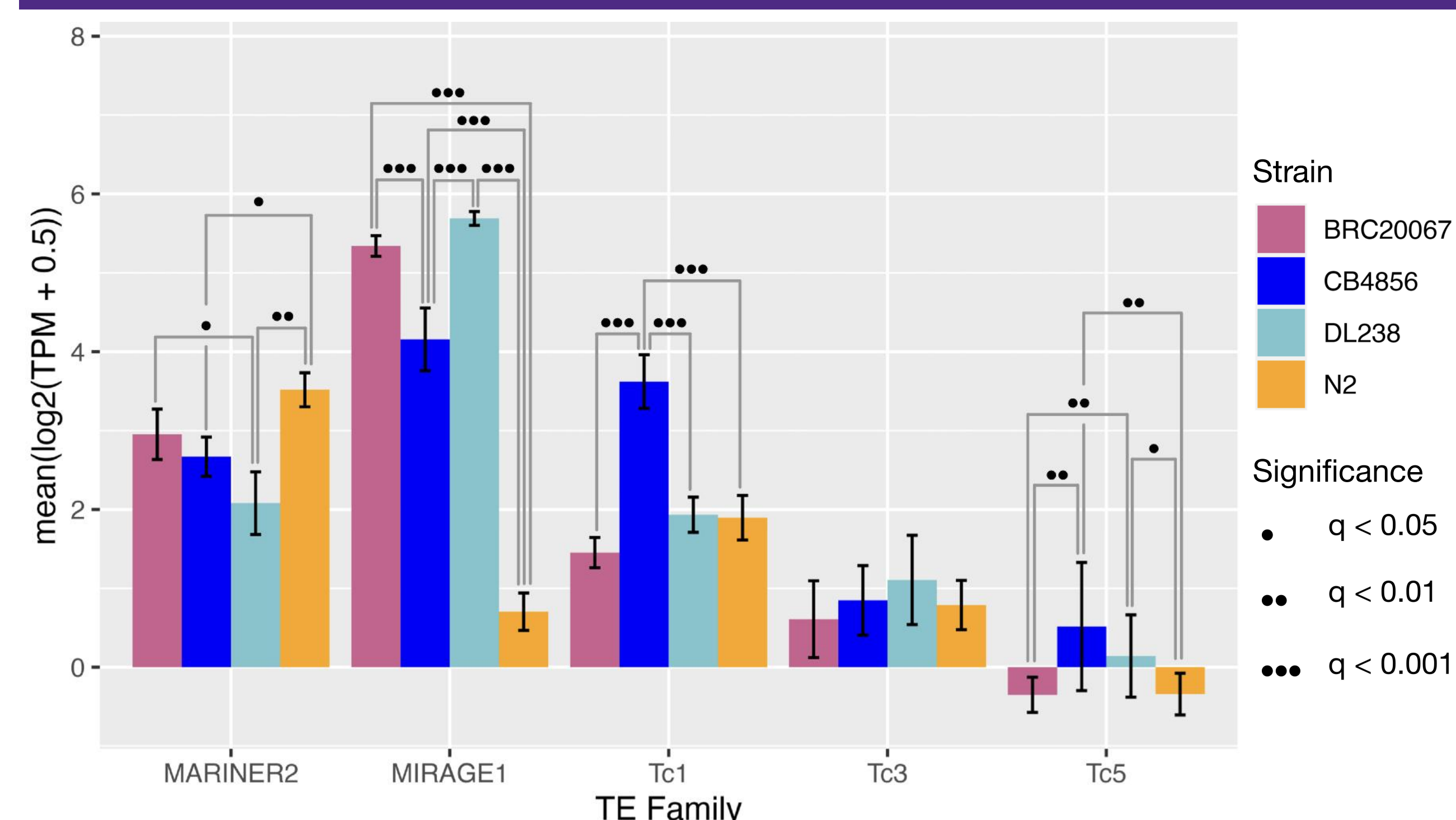


TE Reference Comparison



Log-transformed transcripts per million (TPM) for five TE families across six samples of N2.

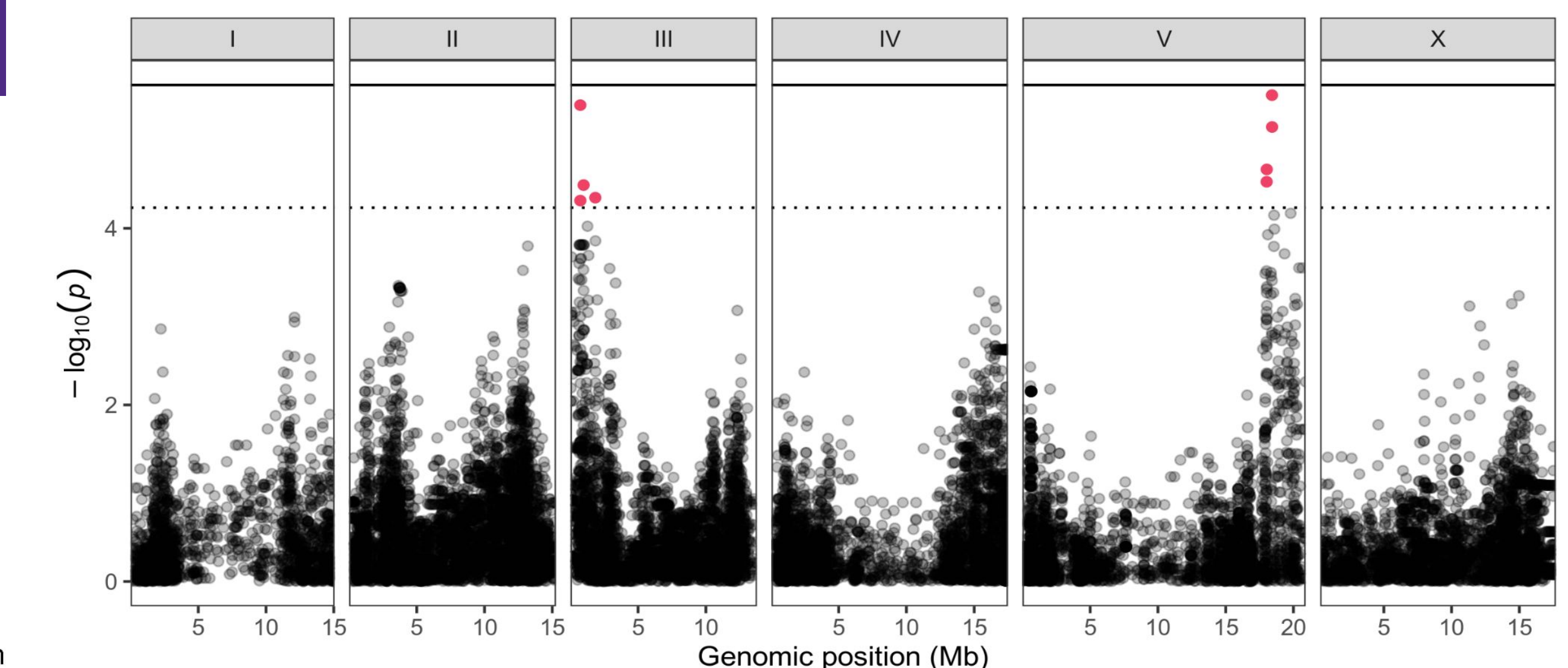
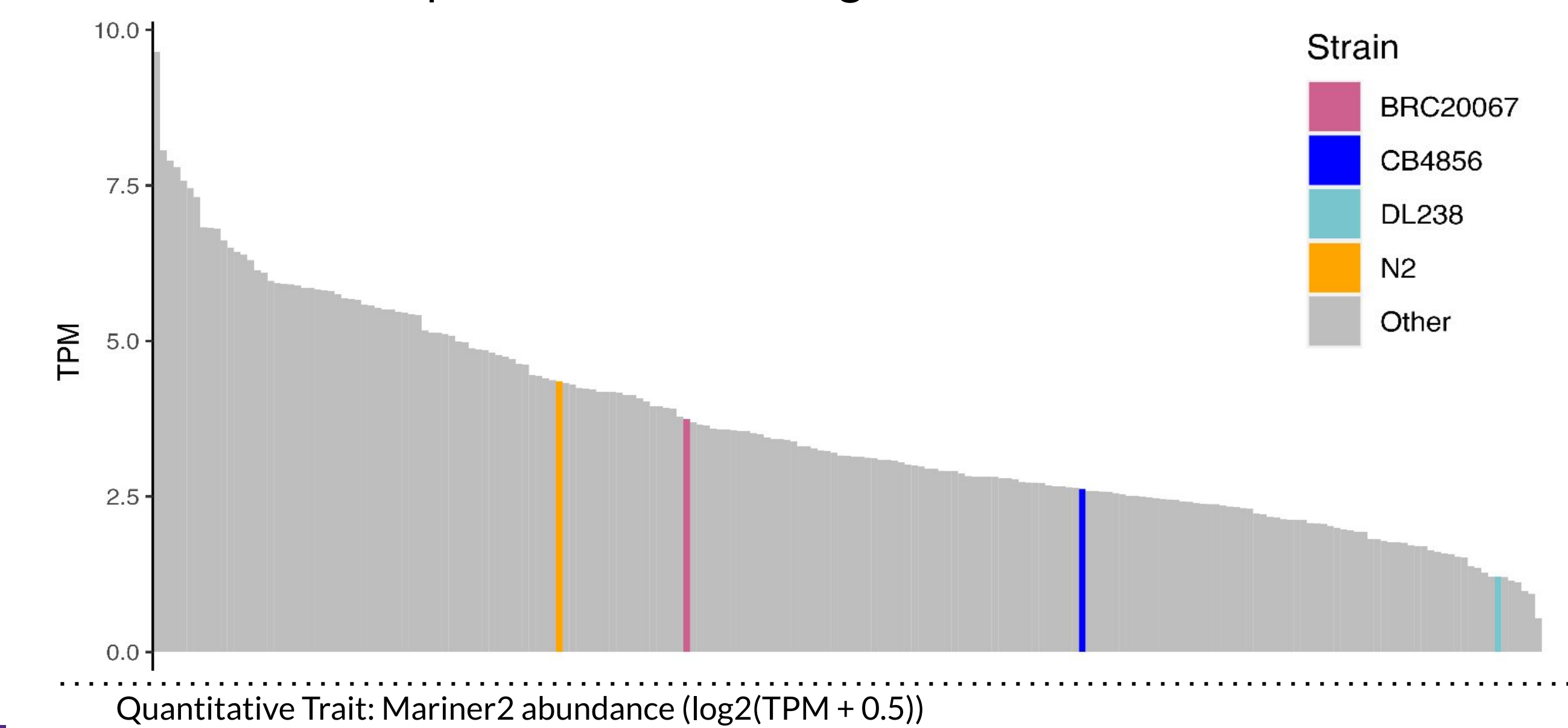
Differential Expression Analysis



Log-transformed TPM for five TE families across four strains (six samples each) quantified with Dfam as reference. Likelihood ratio test q-values (FDR-adjusted p-values) were used to determine significant differences in expression between pairs of strains for each TE family.

Genome-Wide Association Study

TE transcript abundances of 207 *C. elegans* strains were quantified with Dfam sequences as reference. We treated the transcript abundances for each TE family as a quantitative trait. Using NemaScan, we searched for single nucleotide polymorphisms across the genome that may be associated with differences in TE transcript abundances. The figures below show (i) Mariner2 TPM across the 207 strains, and (ii) the significance of association tests between genotypes at each SNP and Mariner2 transcript abundance among the 207 strains.



Conclusion and Next Steps

Our study analyzed *C. elegans* TE sequences procured from the WormBase and Dfam databases. We:

- Developed two new pipelines: **nema_blast** for finding best pairwise alignment matches and **many_multiple_align** for rapid multiple sequence alignment and consensus generation
- Compared reference TE sequence lists through manual matching, pairwise alignment matching, and multiple sequence alignment
- Identified Dfam consensus sequences as most suitable for future quantification of TE transcripts from RNA-seq data
- Demonstrated differential expression analysis and GWA studies using TE transcript abundances as quantitative traits

Future studies should consider using TE-specific transcript quantification tools as well as **many_multiple_align** to generate strain-specific consensus sequences to serve as reference for quantification.

References

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